
The Iris-Stego Laboratory

A plain-language guide to the PhD thesis: secure covert communication
with biometric-derived keys and message-authenticity vetting

[Companion guide to the live experiment website](#)

Overview

This thesis addresses secure covert communication: how to send a secret message so that no one even suspects it exists, while binding the sender's identity and verifying the message is trustworthy.

The system integrates four techniques usually studied separately: iris recognition, image steganography, biometric-keyed encryption, and real-vs-fake message classification.

The Problem & Research Gap

Prior systems treat steganography, encryption, and fake-news detection in isolation from one another.

The gap: there is no unified framework that ties the sender's identity (via their iris) to the encryption key, vets the message's authenticity before hiding it, and does so bilingually (Arabic and English).

The Contribution

- Biometric-derived keying: the encryption key comes from the sender's iris, not a password.
- Multi-layer covert channel: hide the message, encrypt the carrier, then wrap the key.
- Authenticity gate: a classifier judges whether the message is genuine before it is sent.
- Bilingual scope: an Arabic classifier in addition to the English one.
- Comparative empirical evaluation across several ML and deep-learning models.
- Security upgrade: an AES-256-GCM mode with PBKDF2 key derivation alongside the faithful original.

The Pipeline, Step by Step

- Identify the sender from their iris.
- Vet whether the message is authentic or fake.
- Hide the message inside an ordinary image.
- Encrypt the image with the biometric key (faithful RC4 or secure AES-GCM).
- Transmit the package to the receiver.
- Decrypt, extract the message, and verify integrity.

The Datasets

- MMU Iris Database: ~891 eye images from 46 people (152 MB).
- English news dataset: 134,000 rows labelled real/fake (27 MB).
- Arabic news dataset: 4,837 near-balanced rows.
- Iris feature vectors (feat.csv) used as the encryption keys.

Results

- Iris recognition: ~62% accuracy over 45 people (random guessing ~2%).
- English classifier: ~92% accuracy.
- Arabic classifier: ~88% accuracy.

How the Data Reaches the Website

The raw data (3.9 GB) is never uploaded to the website. Training runs once on the research PC, and the learned knowledge is distilled into small model files (~12 MB total).

These models are imported into the website as-is, so it answers instantly when you upload an image or type a

message, without ever needing the original datasets.

How to Use It

Open the website and follow the steps top to bottom: upload an iris, vet the message, hide it, encrypt it, send it, then receive it. Every step shows an explanation and its measurements, and the experiment log can be exported to CSV for the thesis.